



Reference letter

August 1, 2019

To whom it may concern,

It is my immense pleasure to recommend Viacheslav V. Danilov. In my capacity as a director of the Centre for Computational Imaging and Simulation Technologies in Biomedicine (CISTIB) and a professor of School of Computing of the University of Leeds (Leeds, United Kingdom), I came to know Viacheslav very well. I acted as a scientific supervisor during his research stay in the United Kingdom.

Viacheslav has been carrying out research at the CISTIB of the University of Leeds for a period of 6 months. His research is dedicated to a field of medical imaging and an application of machine/deep learning to image processing. Viacheslav applied deep learning techniques using TensorFlow and modern neural networks (MobileNet, ResNet-50, ResNet-101, and NASNet) to a problem of stenosis detection based on coronary angiography. This task was solved with an accuracy of 95%. Additionally, he has solved a task of stenosis scoring having compared the regular classification/regression approach with an approach of multi-task learning. To solve specific tasks, Viacheslav used TensorFlow, Keras, PyTorch, Fast AI, Python, MATLAB, R and Amazon Web Services (EC2 and S3). The work at the CISTIB enabled Viacheslav to make a research output in the form of a scientific paper "Ray-based segmentation algorithm for medical imaging" presented at an ISPRS International Workshop held in Moscow State University (Moscow, Russia). The paper is devoted to a segmentation algorithm for medical images. This algorithm was successfully validated on a cardiac MRI dataset of the University of York and brain tumour dataset of Southern Medical University.

Despite his tight schedule Viacheslav took an English language course at MDA College (Leeds Language Academy) and participated in different laboratory activities such as an away day event in Scarborough and team meetings. Summing up I would say that Viacheslav is an experienced scientist who has an ability to successfully carry out both theoretical and experimental research.

Yours sincerely,

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